

Agent Workflow Benchmark Blueprint v2

This blueprint defines a modern agent workflow benchmarking framework that compares multiple workflow orchestration patterns on a controlled multi-CSV business dataset. The objective is to evaluate workflow effectiveness, reasoning quality, tool usage, latency, cost, and robustness.

Project Objective

Build and benchmark modern AI agent workflow patterns on a structured business data environment using the same tools, tasks, and LLM to evaluate success rate, reasoning quality, workflow efficiency, tool utilization, latency, and cost.

Environment

Multi-CSV business dataset including Customers, Orders, Products, Inventory, Payments, and Revenue tables stored in SQL, CSV, and vector database for business rules and documentation.

Available Tools

SQL query tool, Python analysis tool, CSV reader, vector search, calculator, and document retrieval system.

Task Levels

Level 1: Retrieval tasks such as revenue lookup and product search.

Level 2: Analytical tasks such as trends and segmentation.

Level 3: Multi-step reasoning tasks such as performance explanation.

Level 4: Decision-making tasks such as strategy recommendations.

Workflow Patterns

1. Single-step LLM
2. Tool-using agent
3. Chain workflow
4. Routing workflow
5. Parallel workflow
6. Orchestrator-workers
7. Evaluator-optimizer
8. Manager agent
9. Multi-agent handoff
10. Human-in-the-loop workflow

Evaluation Metrics

Task success rate, tool accuracy, reasoning steps, latency, token cost, retries, and failure rate.

Tech Stack

LangGraph or CrewAI for orchestration, GPT-4 or Claude for LLM, SQLite or PostgreSQL for data, vector DB for knowledge, and DeepEval or LangSmith for evaluation.

Project Structure

agent-workflow-benchmark/
data/
tasks/
agents/
workflows/
tools/
evaluation/
results/
plots/
scripts/
README.md

Execution Plan

- Week 1: Dataset and environment setup.
- Week 2: Workflow implementation.
- Week 3: Advanced agents and evaluation.
- Week 4: Analysis and reporting.

Deliverables

GitHub repository, agent implementations, evaluation framework, results, charts, research report, and blog article.